DS1HW1

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```
34          continue;
35        }
36        if (number > 999999 || number == 0) {
37          std::cout << "### The number must be in [1,99999
38          continue;
39        } else if (number < 0) {
40          std::cout << std::endl;
41          continue;
42        }
43        std::vector<unsigned long long> fib = CalculateFib
44
45        for (int i = 0; i < fib.size(); i++) {
46
47          long long sqrtfib = (long long)(std::sqrt(fib[i]
48          long count_inner_loops = 1;
49          long long factor = 0;
50          while (1) {
51            if (fib[i] % sqrtfib == 0) {
52              factor =  fib[i] / sqrtfib;
53              break;
54            }
55            if (fib[i] % 2 == 1 && sqrtfib % 2 == 1) { //
56              sqrtfib -= 2;
57            } else {
58              sqrtfib -= 1;
59            }
60
```

功能

- 產生 Fibonacci 數列（迭代法）
- 產生 Fibonacci 數列（遞迴法）
- 產生 0~9 的 M 位數組合（Combination）
- 產生 0~9 的 M 位數排列（Permutation）
- 錯誤訊息與防呆

開始 → 顯示選單 → 輸入指令

判斷指令:

0: 結束

1: 迭代法 Fibonacci

2: 遞迴法 Fibonacci

3: 組合

4: 排列

其他: 錯誤提示

執行功能後回到選單

Fibonacci 數列產生

迭代法：用迴圈計算，效率高

遞迴法：用遞迴計算，增進分遞迴思維

顯示每項計算次數

想法：

將1~number的Fibonacci數一個個存進vector陣列裡，然後在一個個找出每個數字差值最小的兩個因數

想法：

[組合]中右邊的數字一定大於左邊的數字

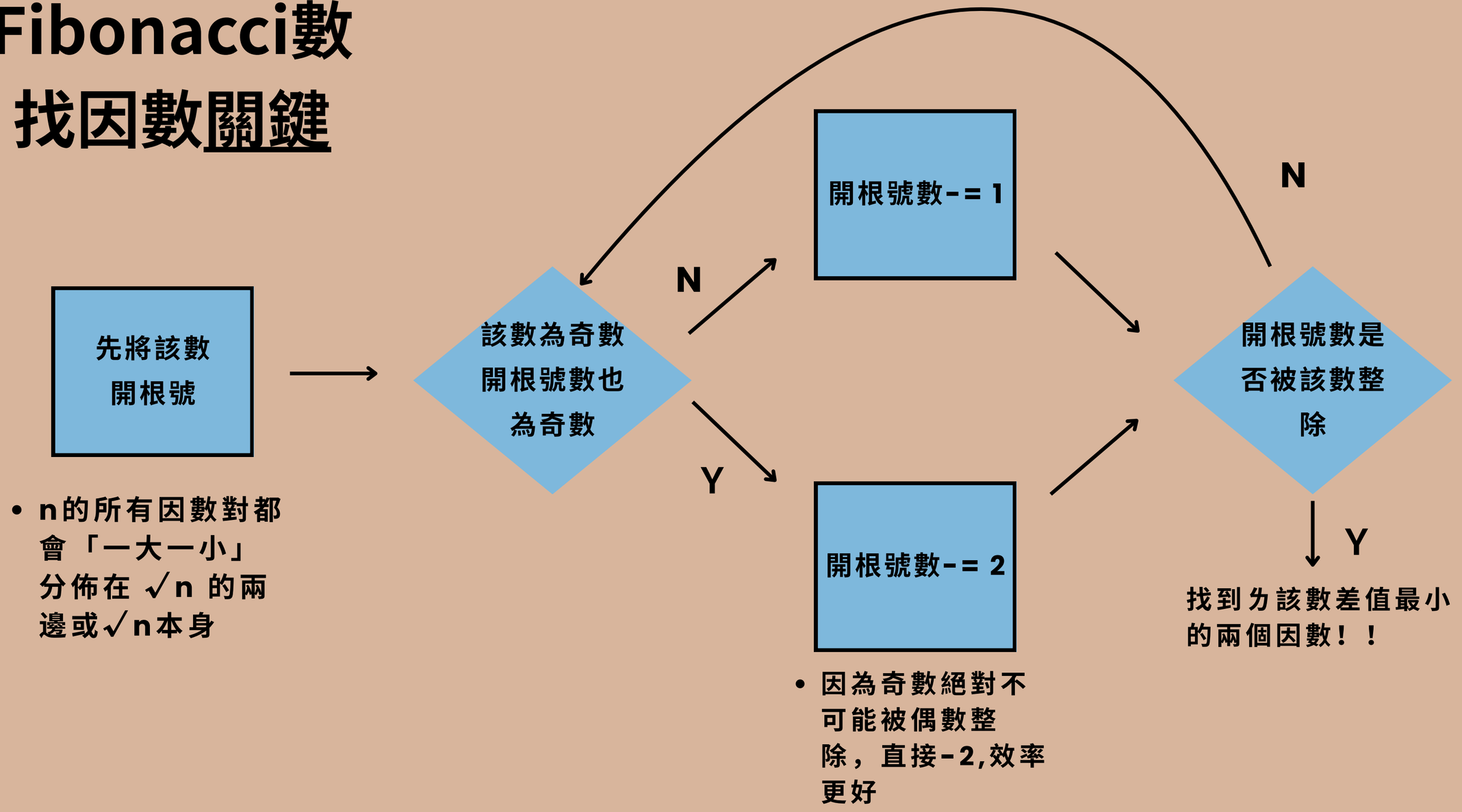
組合/排列產生

組合：不重複、不考慮順序

排列：不重複、考慮順序

顯示所有結果與總數

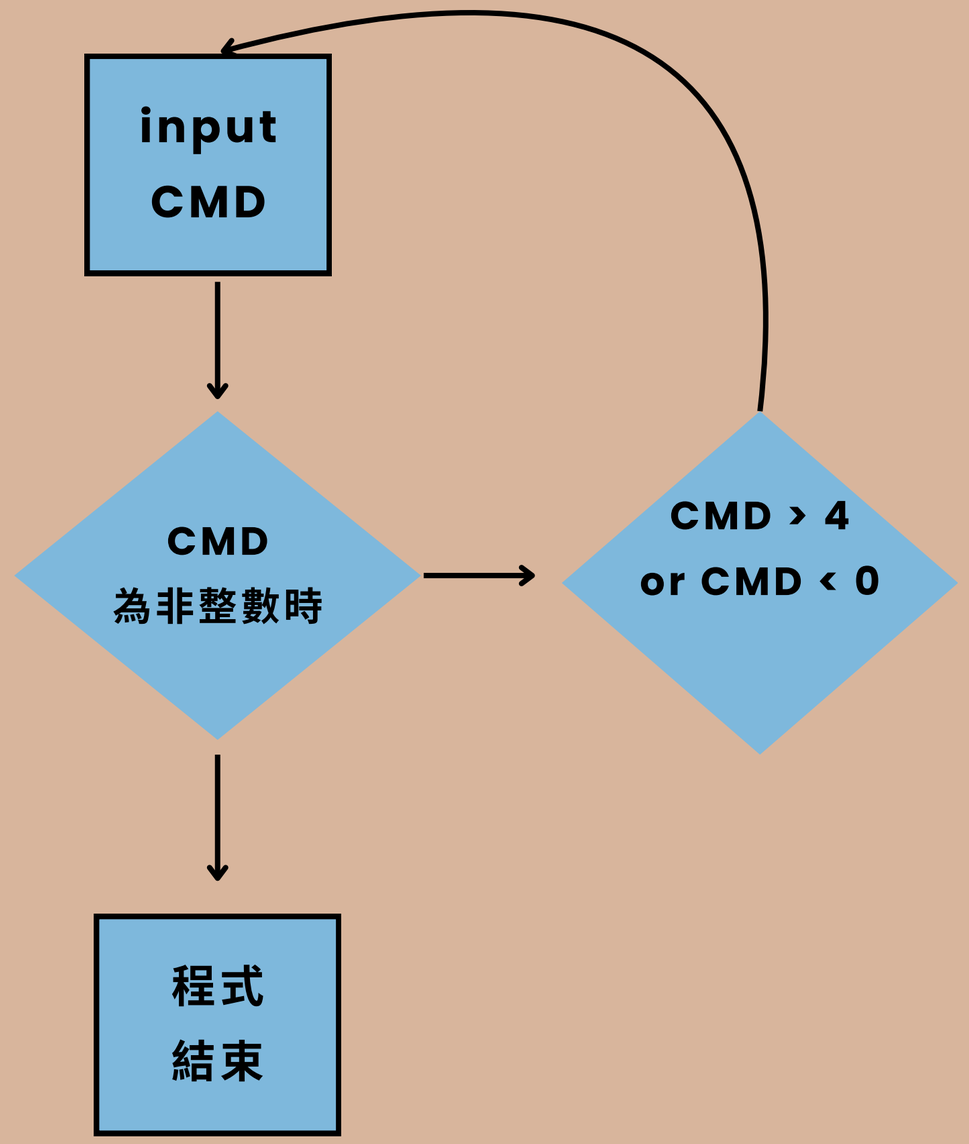
Fibonacci數
找因數關鍵

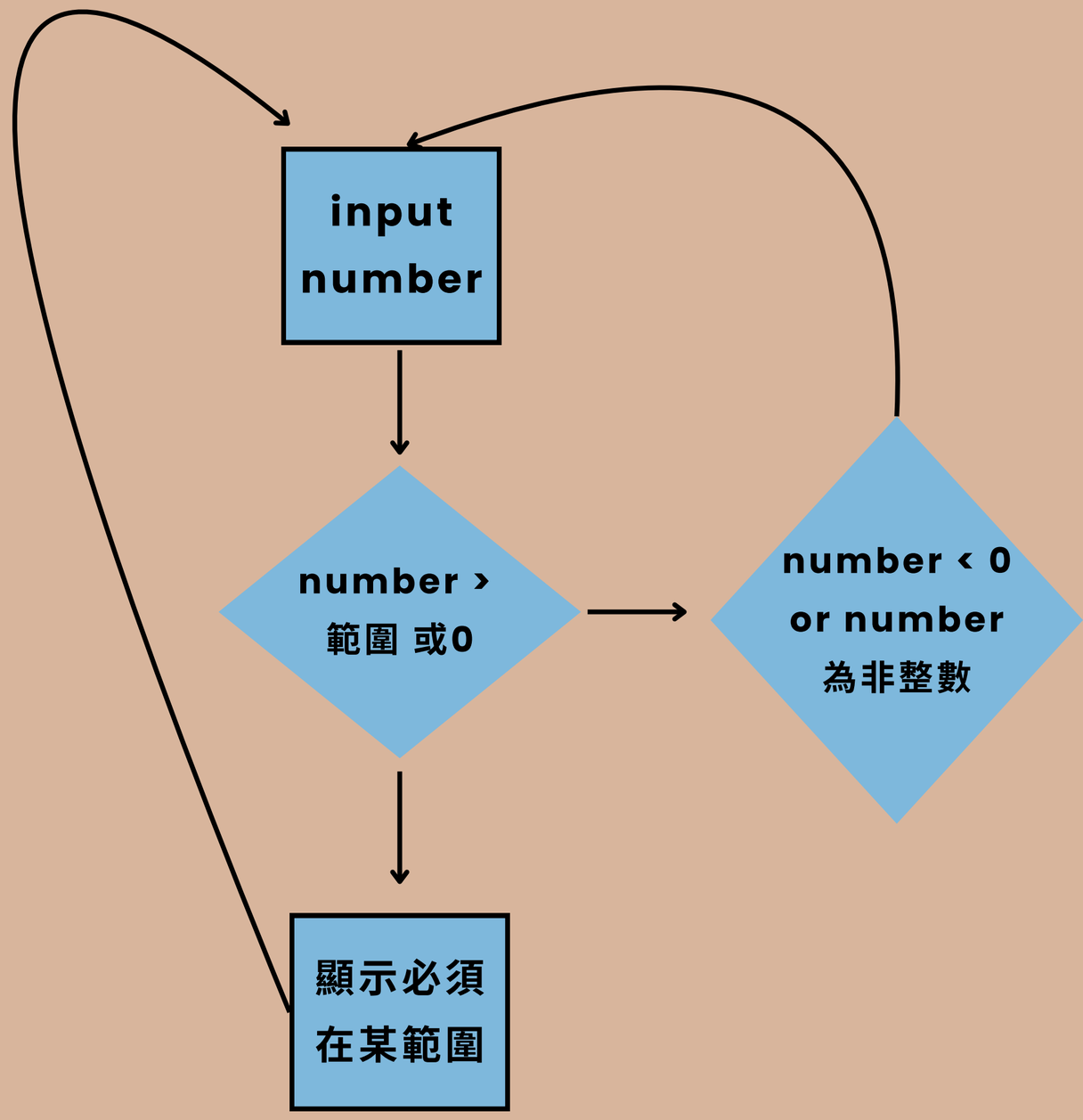


- n的所有因數對都會「一大一小」分佈在 √n 的兩邊或√n本身

- 因為奇數絕對不可能被偶數整除，直接-2,效率更好

錯誤訊息與防呆


input
CMD
CMD
為非整數時
CMD > 4
or CMD < 0
程式
結束


input
number
number >
範圍 或0
number < 0
or number
為非整數
顯示必須
在某範圍

Problem

```
Input a command(0, 1, 2, 3, 4): 2

Input a number: 100
```

```
 [ 50] 20365011074 = 3194 * 6376021      (Inner recursion:          294464 times)
 [ 51] 32951280099 = 121393 * 271443     (Inner recursion:          324530 times)
 [ 52] 53316291173 = 953 * 55945741      (Inner recursion:          439505 times)
 [ 53] 86267571272 = 281656 * 306287     (Inner recursion:          451562 times)
 [ 54] 139583862445 = 294145 * 474541    (Inner recursion:          491294 times)
 [ 55] 225851433717 = 420587 * 536991    (Inner recursion:          518620 times)
zsh: segmentation fault  ./DS1HW1_11327217
```

當我用遞迴產生
Fibonacci 數列時，
遞回超過55次便會發生
Segmentation fault
錯誤訊息

Thanks for your Attention

Date: 20th SEP 2025

DS1HW1_11327217.cpp > main()

```cpp
int main() {
  Operations Operations;
  while (true) {
    g_permutation = 0;
    g_combination = 0;
    int cmd = 0;
    PrintTitle();
    double START, END;
    std::cin >> cmd;
    printf("\n");
    if (cmd == 0) {
      return 0;
    }
    else if (cmd == 1) {
      Operations.IterativeFibonacci();
    } else if (cmd == 2) {
      Operations.RecursiveFibonacci();

    } else if (cmd == 3) {
      START = clock();
      Operations.Combination();
      END = clock();
      std::cout << "T = " <<  (double)clock()/1000 << " ms\n";
    } else if (cmd == 4) {
      START = clock();
      Operations.Permutation();
      END = clock();
      std::cout << "T = " <<  (double)clock()/1000 << " ms\n";

    } else {
      std::cout << "Command does not exist!\n";
    }
    printf("\n");
```